Part 1: L-Attributed SDT for Thompson's Construction

Note: Every S-attributed grammar is inherently a subset of L-attributed grammars. We will use two synthesized attributes for each non-terminal to represent the NFA fragment: `.s` (start state) and `.e` (end state). This perfectly satisfies the L-attributed requirement.

CFG and Semantic Rules

We use non-terminals A (Union), B (Concatenation), C (Kleene Star), and D (Primary) to handle operator precedence.

- **Rule 1 (Char):** $D \to c$
  $D.s = \text{new}()$
  $D.e = \text{new}()$
  $\text{add_trans}(D.s, c.\text{val}, D.e)$
- **Rule 2 (Parentheses):** $D \to (A)$
  $D.s = A.s$
  $D.e = A.e$
- **Rule 3 (Kleene Star):** $C \to D*$
  $C.s = \text{new}()$
  $C.e = \text{new}()$
  $\text{add_trans}(C.s, \epsilon, D.s)$
  $\text{add_trans}(D.e, \epsilon, D.s)$
  $\text{add_trans}(D.e, \epsilon, C.e)$
  $\text{add_trans}(C.s, \epsilon, C.e)$
- **Rule 4 (No Star):** $C \to D$
  $C.s = D.s, \quad C.e = D.e$
- **Rule 5 (Concatenation):** $B \to B_1 C$
  $B.s = B_1.s$
  $B.e = C.e$
  $\text{add_trans}(B_1.e, \epsilon, C.s)$
- **Rule 6 (No Concat):** $B \to C$
  $B.s = C.s, \quad B.e = C.e$
- **Rule 7 (Union):** $A \to A_1 \mid B$
  $A.s = \text{new}()$
  $A.e = \text{new}()$
  $\text{add_trans}(A.s, \epsilon, A_1.s)$
  $\text{add_trans}(A.s, \epsilon, B.s)$
  $\text{add_trans}(A_1.e, \epsilon, A.e)$
  $\text{add_trans}(B.e, \epsilon, A.e)$
- **Rule 8 (No Union):** $A \to B$
  $A.s = B.s, \quad A.e = B.e$

Part 2: Step-by-Step NFA Construction for $a \mid (ab^*a)$

We evaluate the expression bottom-up. Let the input tokens be $c_1 = a$, $c_2 = a$, $c_3 = b$, and $c_4 = a$.

- **Step 1 ($c_1 = a$):** $D_1 \to c_1$
  $D_1.s = q_0, D_1.e = q_1$. Add $\text{trans}(q_0, a, q_1)$.
  (Reduces up to A_1, so $A_1.s = q_0, A_1.e = q_1$)

- **Step 2 ($c_2 = a$):** $D_2 \to c_2$
  $D_2.s = q_2, D_2.e = q_3$. Add trans(q_2, a, q_3).
  (Reduces up to B_2)

- **Step 3 ($c_3 = b$):** $D_3 \to c_3$
  $D_3.s = q_4, D_3.e = q_5$. Add trans(q_4, b, q_5).

- **Step 4 (b^*):** $C_3 \to D_3*$
  $C_3.s = q_6, C_3.e = q_7$.
  Add ϵ-transitions: $(q_6 \to q_4)$, $(q_5 \to q_4)$, $(q_5 \to q_7)$, $(q_6 \to q_7)$.

- **Step 5 (ab^* Concat):** $B_3 \to B_2C_3$
  $B_3.s = q_2, B_3.e = q_7$.
  Add trans$(B_2.e, \epsilon, C_3.s) \implies$ trans(q_3, ϵ, q_6).

- **Step 6 ($c_4 = a$):** $D_4 \to c_4$
  $D_4.s = q_8, D_4.e = q_9$. Add trans(q_8, a, q_9).
  (Reduces up to C_4)

- **Step 7 (ab^*a Concat):** $B_4 \to B_3C_4$
  $B_4.s = q_2, B_4.e = q_9$.
  Add trans$(B_3.e, \epsilon, C_4.s) \implies$ trans(q_7, ϵ, q_8).

- **Step 8 (Parentheses):** $D_5 \to (B_4)$
  $D_5.s = q_2, D_5.e = q_9$.
  (Reduces up to B_5)

- **Step 9 (Union):** $A_3 \to A_1 \mid B_5$
  $A_3.s = q_{10}, A_3.e = q_{11}$.
  Add trans$(q_{10}, \epsilon, A_1.s) \implies$ trans(q_{10}, ϵ, q_0).
  Add trans$(q_{10}, \epsilon, B_5.s) \implies$ trans(q_{10}, ϵ, q_2).
  Add trans$(A_1.e, \epsilon, q_{11}) \implies$ trans(q_1, ϵ, q_{11}).
  Add trans$(B_5.e, \epsilon, q_{11}) \implies$ trans(q_9, ϵ, q_{11}).